Week 14

MICROECONOMIC THEORY
ECON 323 502/503

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- This course (Microeconomic Theory, Econ 323:502/503, Instructor Huiyi Guo) has benefited from comments from earlier students. Your input will also benefit the course in the future.

- Please take 10 minutes to complete the course evaluation at https://tamu.aefis.net. The link is mobile-phone friendly. The link can also be accessed via Canvas under "Course Evaluations" or under "Module 3" (last row). Thank you.

- Class will resume at 11:20 am/12:55 pm.

Review

Cournot duopoly model is a model in which two firms **simultaneously** choose **the quantity** to produce.

The Nash equilibrium in a Cournot duopoly game is the **Cournot equilibrium**.

An Asymmetric Cournot Model

In a Cournot duopoly market with market demand function $Q(p)=1500-5p$, firm 1 has cost function $C_1(q_1)=120q_1$ and firm 2 has cost function $C_2(q_2)=150q_2$. Notice that the two firms have different cost functions. In the Cournot equilibrium, what are the output choices (firm 1's output, firm 2's output)?

Q(p)=1500-5p. MC for firm 1 is 120. MC for firm 2 is 150.

Step 1(1):
The inverse demand function

Step 1(2):
Revenue of firm 1

Step 1(3):
Marginal revenue of firm 1 MC for firm 1

Step 2:
Setting marginal revenue equal to marginal cost for firm 1

$Q(p)=1500-5p$. MC for firm 1 is 120. MC for firm 2 is 150.

Step 1(1):
The inverse demand function is $p(q)=300-0.2q=300-0.2q_1-0.2q_2$.

Step 1(2):
Revenue of firm 1 is $p(q_1+q_2)q_1=(300-0.2q_1-0.2q_2)q_1 =300q_1-0.2q_1^2-0.2q_2q_1$.

Step 1(3):
Marginal revenue of firm 1 is $300-0.4q_1-0.2q_2$. MC for firm 1 is 120.

Step 2:
Setting marginal revenue equal to marginal cost for firm 1

Q(p)=1500-5p. MC for firm 1 is 120. MC for firm 2 is 150.

Step 1(1):
The inverse demand function

Step 1(2):
Revenue of firm 2

Step 1(3):
Marginal revenue of firm 2 MC for firm 2

Step 2:
Setting marginal revenue equal to marginal cost for firm 2

$Q(p)=1500-5p$. MC for firm 1 is 120. MC for firm 2 is 150.

Step 1(1):
The inverse demand function is $p(q)=300-0.2q=300-0.2q_1-0.2q_2$.

Step 1(2):
Revenue of firm 2 is $p(q_1+q_2)q_2=(300-0.2q_1-0.2q_2)q_2 =300q_2-0.2q_1q_2-0.2q_2^2$.

Step 1(3):
Marginal revenue of firm 2 is $300-0.2q_1-0.4q_2$. MC for firm 2 is 150.

Step 2:
Setting marginal revenue equal to marginal cost for firm 2, we have $300-0.2q_1-0.4q_2=150$. Isolate q_2, we have $q_2=375-0.5q_1$. (2)

$q_1=450-0.5q_2$. (1)

$q_2=375-0.5q_1$. (2)

Plug (1) into (2), we have $q_2=375-0.5(450-0.5q_2)=375-225+0.25q_2=150+0.25q_2$.

Solve for $q_2=200$.

Plug $q_2=200$ into (1) $q_1=450-0.5*200=350$.

In sum, the Cournot equilibrium is (350, 200).

Surprising or not? Not really, Firm 1 has an advantage because of lower cost.

Bertrand Duopoly

Bertrand duopoly model is a model in which two firms **simultaneously** choose **prices** to post.

The Nash equilibrium in a Bertrand duopoly game is the **Bertrand equilibrium**.

- A Bertrand equilibrium is a pair of prices (p_1, p_2) so that:
  - given Firm 2 charges p_2, Firm 1's best response is p_1;
  - given Firm 1 charges p_1, Firm 2's best response is p_2.

Bertrand Duopoly

- More details:
- Firm 1's cost function of producing q_1 unit of output is $C_1(q_1)$
- Firm 2's cost function of producing q_2 unit of output is $C_2(q_2)$
- Market demand under price p is denoted by Q(p)
- Consumers only buy from the firm with the lowest price.
- Firms share the market if prices are identical.

Example

Suppose $Q(p)=1000-2p$, $C_1(q_1)=200q_1$ and $C_2(q_2)=200q_2$. What is the Bertrand equilibrium?

Discussion: Four cases (assume without loss of generality that $p_1 \geq p_2$)

Case 1: $p_1 > p_2 > 200$.

- Firm 1 earns zero profit.
- Firm 1 can charge a price slightly lower than p_2 and earn a positive profit.

Case 2: $p_1 > p_2 = 200$.

- Firm 2 earns zero profit.
- Firm 2 can slightly increase the price and earn a positive profit.

Case 3: $p_1 = p_2 > 200$.

- Firms share the market and earn positive profit.
- Firm 1 (or 2) can decrease price slightly, gets the entire market, and benefit.

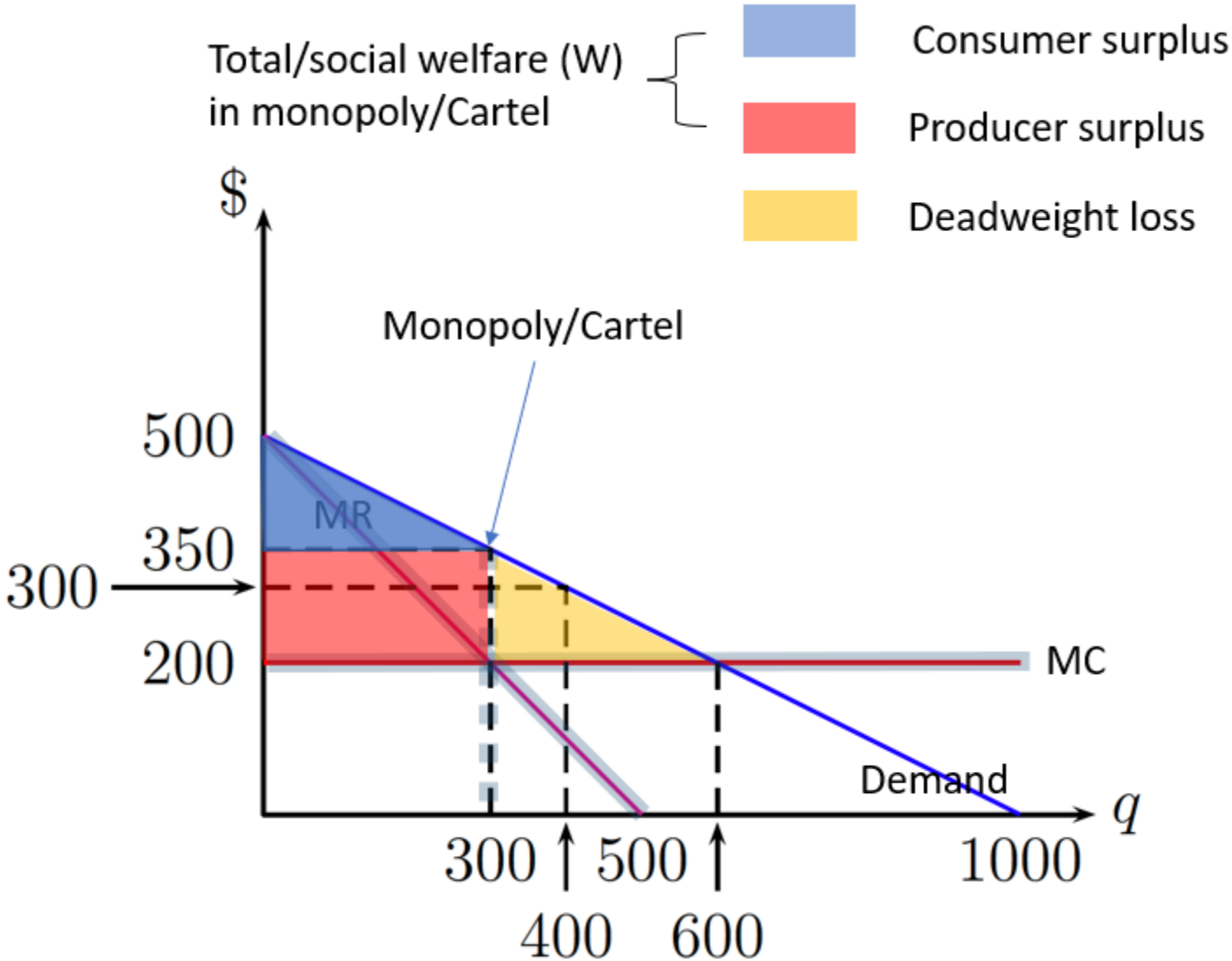
Case 4: $p_1 = p_2 = 200$. ← Bertrand equilibrium.

- Firms share the market and earn zero profit.
- If a firm decreases price, will suffer from a loss. If a firm increases price, will lose the entire market. No profitable deviation.

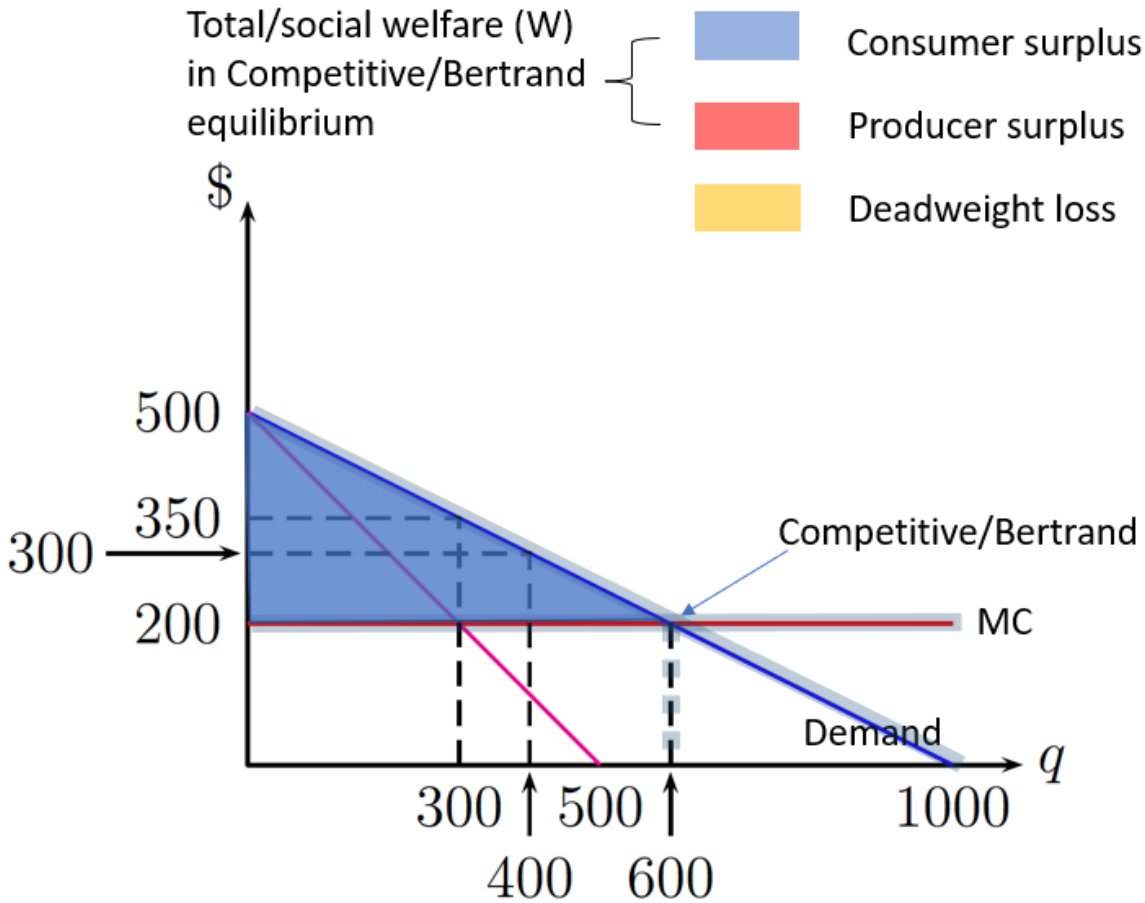
Summary of Symmetric Bertrand equilibrium

- Firm 1 and 2 post prices $p_1 = p_2 = 200$ (marginal cost, average cost).
- Both firms make zero profit.
- Firms share the market: total output Q(200)=1000-2*200=600.
- The price/total output level/total profit are identical with those under competitive equilibrium.

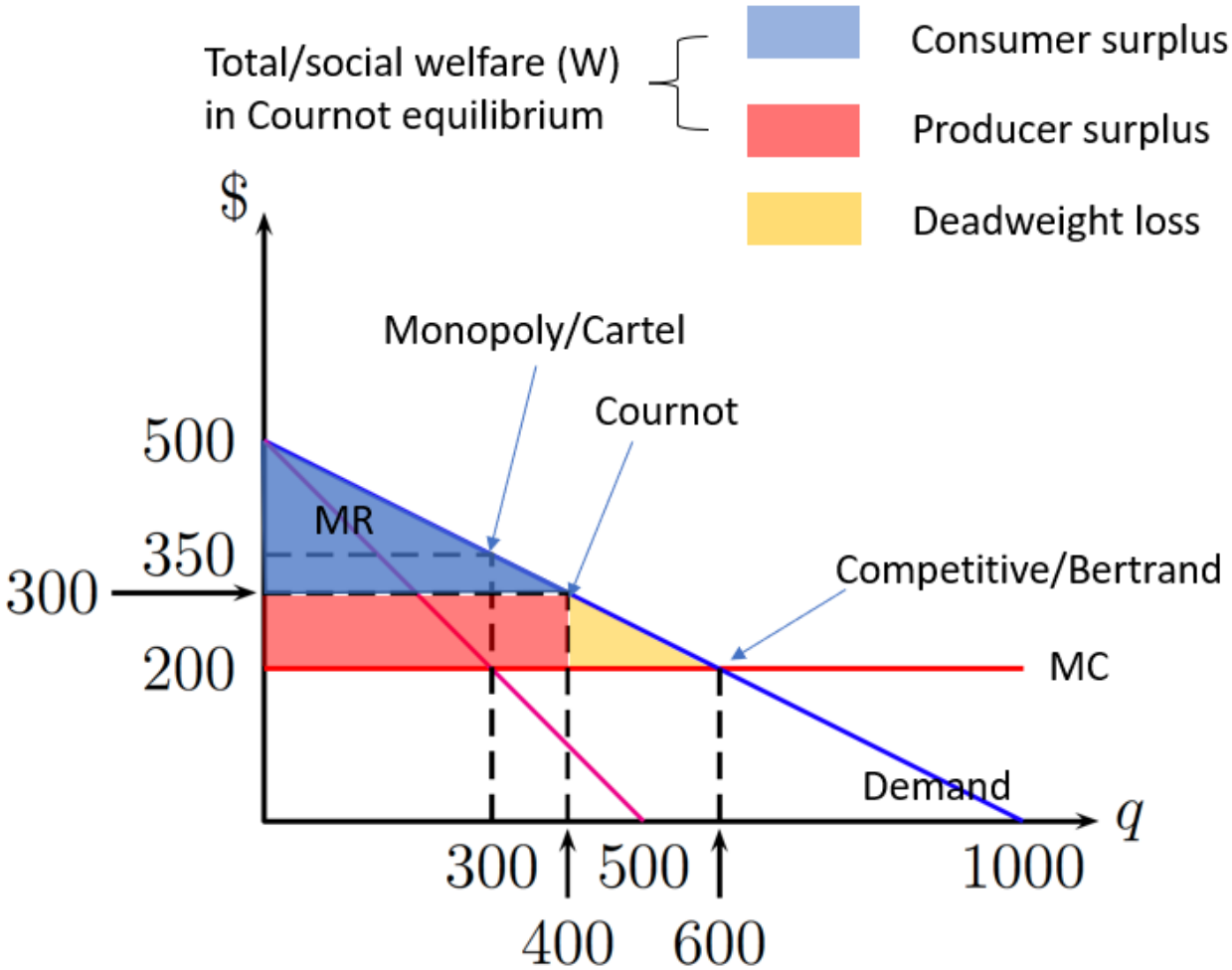
Graph: Monopoly Welfare



Graph: Bertrand Welfare



Graph: Cournot Welfare



Comparison

	Monopoly /Cartel	Cournot	Competitive /Bertrand
Total Output	lowest →		highest
Price	highest →		lowest
Total Profit	highest →		lowest
Total Welfare	lowest →		highest

Extension: Bertrand Model with Discrete Prices

- Prices restricted to \$0.01 increments.
- $Q(p)=1000-2p$, $C_1(q_1)=200q_1$ and $C_2(q_2)=200q_2$.

Is (200, 200) a Nash equilibrium?

Is (200.01, 200.01) a Nash equilibrium?

Is (200, 200.01) a Nash equilibrium?

Is (201, 201) a Nash equilibrium?

Key information: MC=200 for both firms.

The nature of Bertrand equilibrium is Nash equilibrium.

Nash equilibrium requires that each player plays a best response to the other player's strategy.

Is (200, 200) a Nash equilibrium?

- Yes. Both firms play a best response.

Is (200.01, 200.01) a Nash equilibrium?

- Yes. Both firms play a best response.

Is (200, 200.01) a Nash equilibrium?

- No. Firm 1 does not play a best response.

Is (201, 201) a Nash equilibrium?

- No. Undercutting is a profitable deviation. (Verify on your own!!!)

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